

CUSTOMER FEATURE GUIDE

Aether

Universal runtime portability.

Deploy once, then move workloads across Podman, Kubernetes, KubeVirt, and Metal3 from a single YAML spec — no infrastructure rewrites.

by ZyvorAI Labs

What is Aether?

Aether is a universal runtime control plane: one workload spec describes what to run, and Aether deploys it to the right runtime, explains why, and migrates it between runtimes with production strategies. It ships a Rust CLI, an interactive TUI, and a React web dashboard over a shared control plane — think of it as Terraform for where your workloads run, not just what they are.

4

Runtimes (Podman, K8s, KubeVirt, Metal3)

16

Migration paths

5

Migration strategies

65+

CLI commands

What's inside this guide

1 · One Spec, Four Runtimes	7 features
2 · Production Migration Engine	7 features
3 · Intent Decision Engine	7 features
4 · Discovery, Assessment & Cloud Exit	7 features
5 · Operations & Lifecycle	7 features
6 · Observability & Cost Insight	7 features
7 · AI Ops & Autonomy	7 features
8 · Security, Policy & Compliance	7 features
9 · Confidential Computing (Ragnarok)	6 features
10 · Integrations & Extensibility	8 features
Getting started & support	→

One Spec, Four Runtimes

A single Aether workload spec validates and deploys everywhere — from a laptop container to bare metal.

Runtime	Best for
Podman	Local dev and edge containers
Kubernetes	Cluster orchestration, services, scaling
KubeVirt	VM isolation and GPU passthrough
Metal3	Bare-metal performance



Universal Workload Spec

Describe a workload once in the aether/v1 YAML schema — image, resources, ports, persistence, secrets, and intent.

Stop maintaining four different manifest dialects for the same app.



Validate Before Deploy

Check spec syntax, intent, and per-runtime compatibility before anything ships.

Catch misconfiguration at author time, not in production.



Run Anywhere

Deploy a spec to Podman, Kubernetes, KubeVirt, or Metal3 — with the runtime auto-selected or pinned by flag.

The same command targets local dev, clusters, VMs, and bare metal.



Podman & Local Dev

Run containers locally or at the edge through the Podman adapter with the identical spec you ship to prod.

True dev/prod parity from the first line of YAML.



Kubernetes Native

Generate Deployments, Services, Ingress, HPA, ConfigMaps, and Secrets from one spec via the Kubernetes adapter.

Full cluster orchestration without hand-writing manifests.



KubeVirt VMs

Deploy the workload as a KubeVirt virtual machine for strong isolation and GPU passthrough.

Move a container to a VM when you need harder boundaries — no rewrite.



Metal3 Bare Metal

Target bare-metal hosts through the Metal3 adapter for maximum performance.

Reach dedicated hardware from the same control plane as everything else.

Production Migration Engine

Move a running workload between any two runtimes with health gates, connection draining, and automatic rollback.



16 Migration Paths

Migrate across every ordered pair of the four runtimes with a single migrate command.

[Escape a legacy VM or container stack without a re-platforming project.](#)



Blue-Green Cutover

Stand up the target (green) while the source (blue) keeps serving, health-gate it, then cut over.

[Zero-downtime moves that only switch traffic once the target is proven healthy.](#)



Rolling & Immediate

Choose incremental rolling steps, an immediate stop-then-start, or canary — per migration.

[Match the migration risk profile to each workload.](#)



Automatic Rollback

On a failed health gate the engine cleans up the target and restores the source automatically.

[A failed migration leaves you where you started, not half-migrated.](#)



Health Gates & Draining

Exponential-backoff health validation and connection draining (up to a 30s cap) guard every cutover.

[In-flight requests finish before the old instance goes away.](#)



Migration Trace

Stream every phase — snapshot, build-target, health-gate, drain, update-state — to stderr for debugging.

[See exactly where a migration is and why, step by step.](#)



AI Migration Advice

Get an AI assessment of moving a named workload to a target runtime before you commit.

[Understand the risks and gains of a move up front.](#)

Migration is designed for stateless and container-image portability. Persistent volume data is not automatically carried across runtimes — see caveats.

Intent Decision Engine

Score and rank runtimes on cost, performance, reliability, and availability — with reasons, not a black-box pick.



Explainable Placement

aether decide ranks every eligible runtime with per-dimension scores and human-readable reasons.

[Stop guessing which runtime fits — see the math behind the choice.](#)



Four-Dimension Scoring

Each runtime is scored 0–1 on cost, performance, reliability, and availability with configurable weights.

[Tune the ranking to your organization's priorities.](#)



Intent Multipliers

An intent block (goal, SLA, budget, resilience, compliance) reshapes the weights toward your objective.

[Encode 'low-latency' or 'cost-optimized' once and let the engine honor it.](#)



Workload Classification

Specs are auto-classified (stateless, stateful, GPU, bare-metal, batch) to filter ineligible runtimes.

[GPU jobs never get scored onto runtimes that can't run them.](#)



Runtime Compare

Side-by-side comparison of cost, capabilities, and limitations across all four runtimes for a spec.

[One view to justify a placement decision to your team.](#)



Intent Debugger (Radar)

The dashboard renders scores as a radar chart so you can see trade-offs at a glance.

[Turn placement scoring into a picture stakeholders understand.](#)



Affinity Learning

Recommend runtimes per workload class from a learned compatibility matrix and usage statistics.

[Recommendations sharpen as the platform learns your fleet.](#)

Discovery, **Assessment** & Cloud Exit

Connect to existing clusters, inventory their apps, score portability, and generate a cloud-exit plan.



Cluster Connections

Save and test connections to EKS, AKS, GKE, OpenShift, Rancher, Tanzu, k3s, RKE2, Podman, or Compose.

[Point Aether at what you already run — no agents to install first.](#)



Application Discovery

Scan a connected cluster and persist an inventory snapshot of its applications and workloads.

[Get a real map of what's running before you plan a move.](#)



Dependency Graph

Surface the dependency graph for a discovered application across its services and data stores.

[See what breaks together before you migrate anything.](#)



Portability Assessment

Assess an application's migration readiness and complexity, with an optional target-cluster preflight.

[Know which apps are easy wins and which need work.](#)



Migration Plan CRD

Generate a MigrationPlan (aether.zyvor.dev/v1alpha1) with a recommended or overridden strategy.

[Turn an assessment into a reviewable, versioned plan artifact.](#)



Cloud Exit Report

Produce a Cloud Exit Assessment report for a connection in Markdown, JSON, or text.

[A shareable, evidence-backed case for leaving a cloud.](#)



Move (Shadow Deploy)

Mirror images and shadow-deploy an app to a target with no external traffic, then cut over or roll back.

[Rehearse a cross-cluster move safely before flipping DNS.](#)

Operations & Lifecycle

Day-2 workload operations — inspect, scale, restart, back up, roll back, and drain nodes from one tool.



Logs, Exec & Port-Forward

Tail logs with follow mode, exec into containers, forward ports, and copy files across runtimes.

[The kubectl toolbox for every runtime, in one CLI.](#)



Scale & Restart

Scale a Kubernetes workload to a target replica count or trigger a rolling restart.

[Routine scaling actions without leaving Aether.](#)



Health-Aware Orchestration

Register workloads for health monitoring, watch at an interval, and trip or reset circuit breakers.

[Continuous health signal that feeds migrations and auto-rollback.](#)



Backup & Restore

Snapshot workload state locally or to a remote endpoint and restore or merge it back.

[State you can recover after a bad change or a lost host.](#)



Snapshot Rollback

Roll a workload back to any listed snapshot version of its stored state.

[Undo a deployment to a known-good point in seconds.](#)



Node Maintenance

Cordon, uncordon, and safely drain Kubernetes nodes for maintenance windows.

[Take hardware offline without evicting workloads by hand.](#)



Watch & Auto-Redeploy

Watch a spec file and auto-redeploy on change, or reconcile a whole directory of specs.

[A fast inner loop that keeps live state matching your files.](#)

Observability & Cost Insight

A k9s-level view of the fleet — metrics, events, audit, drift, cost, and SLA — across CLI, TUI, and web.



Web Dashboard

A 19-page React dashboard with SSE real-time updates, command palette (⌘K), and a live runtime-fabric topology graph.

[One pane of glass for the whole runtime fleet.](#)



Interactive TUI

A k9s-style terminal dashboard for real-time monitoring, logs, and workload navigation.

[Full situational awareness without leaving the terminal.](#)



Prometheus Metrics

Export workload and control-plane metrics in Prometheus format for your existing stack.

[Plug Aether straight into Grafana and Alertmanager.](#)



Events & Audit Trail

Filterable event stream by severity plus a tamper-evident audit trail of every action.

[Know what changed, when, by whom — with integrity you can prove.](#)



Drift Detection

Detect configuration drift between spec, stored state, and live runtime — with optional auto-reconcile.

[Silent config drift becomes a visible, fixable signal.](#)



Cost Estimation

Estimate workload cost across AWS, Azure, GCP, DigitalOcean, and Linode with spot/reserved discounts.

[Compare the price of a workload before you place it.](#)



SLA Compliance

Set SLA tiers per workload and check observed uptime, latency, error rate, and restarts against them.

[Turn availability promises into monitored, reportable targets.](#)

SECTION 7

AI Ops & Autonomy

Ask Zeus, the terminal and dashboard AI assistant, plus an intelligence layer that predicts, heals, and advises.



Ask Zeus Copilot

An interactive AI ops assistant in the terminal and a permanent dashboard rail for natural-language questions.

[Ask the platform what's wrong and what to do about it.](#)



Predictive Scaling

Recommend scaling actions and forecast capacity from observed workload behavior.

[Size ahead of demand instead of reacting to alerts.](#)



Self-Healing & Remediation

An autonomy layer detects anomalies and proposes or runs remediation for failing workloads.

[Common failures get fixed before they page a human.](#)



Log Anomaly Analysis

Analyze a workload's logs for anomalies and recurring patterns.

[Find the needle in noisy logs without a query language.](#)



Digital Twin

Model the fleet as a digital twin to reason about capacity, placement, and change impact.

[Test a decision against a model before touching production.](#)



Command Center Briefing

A narrative fleet briefing summarizing health, intent, SLA, and the next recommended actions.

[Start the day with a plain-language state of the fleet.](#)



Autonomous FinOps & SRE

Agent panels for autonomous placement, FinOps budget enforcement, and SRE reliability workflows.

[Delegate routine cost and reliability toil to guided agents.](#)

AI features connect to an OpenAI-compatible LLM or the Forge inference gateway; several autonomy panels ship as opt-in Labs previews.

Security, Policy & Compliance

Encrypted secrets, enterprise SSO, RBAC, policy gates, and supply-chain evidence built into the control plane.



AES-256-GCM Secrets

Encrypt all secret values at rest with authenticated AES-256-GCM, with a managed rotation lifecycle.

Secrets stay protected on disk and rotate on schedule.



Enterprise SSO

Authenticate via LDAP, SAML, or OIDC with role mapping from your identity provider.

Plug Aether into the SSO your organization already runs.



RBAC & Audit Export

Role-based access control across API and UI, with exportable audit logs for compliance.

Least-privilege access with an exportable paper trail.



Policy Gate on Deploy

Enforce production/development policy sets — or a custom OPA/Rego policy — before a workload ships.

Non-compliant workloads are blocked at deploy, not audited after.



SBOM Export & Verify

Produce and validate CycloneDX software bills of materials for workloads.

Supply-chain evidence auditors and customers can verify.



TLS Lifecycle

Serve the API over HTTPS with self-managed certificate regeneration and opt-in auto-renewal.

Encrypted endpoints without manual cert babysitting.



Hardened Control Plane

API key auth, CORS protection, rate limiting, input validation, atomic state locking, and hardened file permissions.

Production-safe defaults across the whole surface.

Confidential Computing (Ragnarok)

Run and migrate attested, measured workloads on trusted execution environments with sovereign-cloud controls.



Measured Image Catalog

Sign VM disk images into a verified catalog and verify a file hash or launch digest against it.

[Only known, signed images ever launch.](#)



Attestation & Trust

Verify TEE attestation verdicts (SEV-SNP, TDX) and gate placement on the trust result.

[Prove a workload runs on genuine confidential hardware.](#)



Tenant Isolation Check

Validate a workload spec against tenant isolation policy before scheduling.

[Catch isolation violations before a tenant is exposed.](#)



GuestKit Offline Inspection

Inspect a VM image offline in pre-launch, offline-policy, post-shutdown, or attested-repair modes.

[Assurance on a VM's contents without booting it.](#)



Confidential Migration

Plan and track encrypted, attested migration of confidential workloads (confidential blue-green).

[Move sensitive VMs between runtimes without breaking the trust chain.](#)



Sovereign Compliance Check

Check a workload against sovereign-cloud residency and compliance constraints.

[Keep regulated workloads inside the boundaries they require.](#)

Confidential features require compatible TEE hardware (AMD SEV-SNP or Intel TDX) and a Kata/KubeVirt confidential runtime.

Integrations & Extensibility

Slot Aether into your platform — GPU, storage, GitOps, packaging, plugins, edge fleets, and multi-tenant hosting.



Forge GPU/AI

Read live GPU capacity, node inventory, placement recommendations, and cost from the Forge control plane.

[Reason about GPU/AI infrastructure in the same tool as everything else.](#)



Atlas Storage

Provision policy-driven persistent volumes (Ceph, NFS, ZFS) through Atlas with snapshot and clone support.

[Intent-based storage instead of hand-rolled PVCs.](#)



GitOps Reconcile

Bind a Git repo and branch, sync manually or auto-reconcile on a poller in serve mode.

[Declarative, Git-sourced fleet state with drift correction.](#)



Helm & Package Deploy

Export any workload as a Helm chart, or install via Docker images and deb/rpm packages.

[Deliver Aether-managed workloads through your existing pipeline.](#)



Plugin System

Discover and register runtime plugins from manifests to extend Aether beyond the built-in adapters.

[Add new runtimes and behaviors without forking the core.](#)



Webhooks & Notifications

Route severity-filtered notifications to webhook channels with a retry queue and test/flush controls.

[Wire fleet events into Slack, PagerDuty, or any endpoint.](#)



Edge Fleet & Federation

Register edge sites that drain a reconcile queue from the control plane, with fleet federation across clusters.

[Manage disconnected and edge sites from one control plane.](#)



Hosted Multi-Tenancy

A hosted control-plane foundation with tenant registry, switcher, managed upgrades, and billing/metering.

[Run Aether as a multi-tenant service for many teams.](#)

GETTING STARTED

Your first 30 minutes

A suggested path to value on day one.

- 1 Build & initialize**
Clone the repo, run `cargo build --release`, then `aether init` to run the first-time setup wizard.
- 2 Author a spec**
Start from a template (`aether template rest-api`) or an example workload, then `aether validate --spec workload.yaml`.
- 3 Decide & deploy**
Run `aether decide --spec workload.yaml --explain` to see the ranked runtime, then `aether run --spec workload.yaml`.
- 4 Migrate with confidence**
Move a live workload with `aether migrate my-app kubevirt --strategy blue-green`, with rollback on failure.
- 5 Open the dashboard**
Run `aether serve` and browse `http://localhost:5090` for the web UI, Command Center, and Ask Zeus copilot.

Support & contact

Aether is developed by **ZyvorAI Labs**. For onboarding, a proof-of-concept, or a guided demo, contact your account team at **info@zyvor.dev**.

Good to know: Aether's portability guarantees are strongest for stateless and container-image workloads: migration rebuilds the app image for the target runtime and updates workload state atomically, but persistent volume data is not automatically carried across runtimes and blue-green connection draining is capped at ~30s. Integrations are optional and mostly read-only — Forge (GPU) and Atlas (storage) are off unless configured, and the hosted multi-tenant control plane ships today as a foundation with fleet federation still maturing. Confidential computing requires compatible TEE hardware (AMD SEV-SNP / Intel TDX) and a Kata/KubeVirt confidential runtime, and several AI autonomy panels are opt-in Labs previews that depend on an external or Forge-hosted LLM.

Aether

Deploy once, then move workloads across Podman, Kubernetes, KubeVirt, and Metal3 from a single YAML spec — no infrastructure rewrites.

info@zyvor.dev · zyvor.dev